COLLEGE PROJECT PRESENTATION

Flight Delay Prediction via Gradient Boosting and Temporal Deep Learning

Phase 1: data analysis, feature engineering, Logistic Regression, XGBoost and evaluation

Five presenter roles

- Data Analysis
- Feature Engineering
- Baseline Model
- XGBoost
- Evaluation & Docs

Problem and Objective

Presenter: Evaluation & Docs

- Delays cost time and money for passengers and airlines
- Goal: predict before departure if a flight arrives 15+ minutes late
- Task type: binary classification (delayed or not delayed)
- Phase 1 compares Logistic Regression (baseline) with XGBoost

Phase 1 scope: analysis, features, two models, evaluation

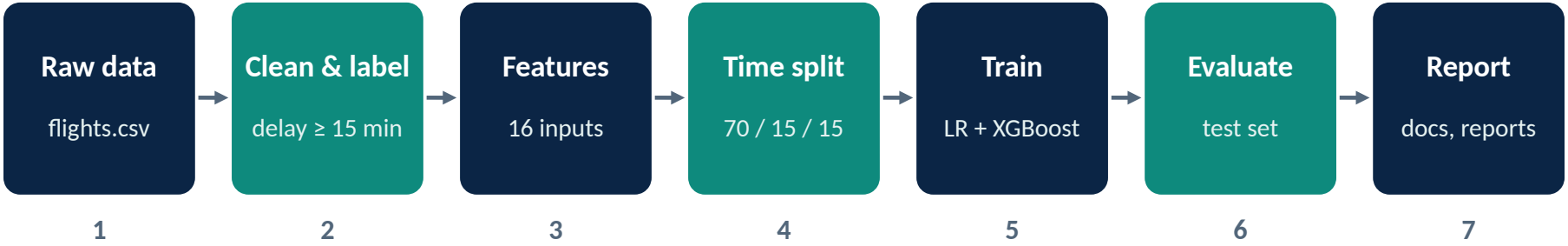
336,776
flights in the raw dataset (2013)

24.47%
of labelled flights arrive 15+ min late

16
pre-departure input features

Project Workflow

Presenter: Data Analysis



Done in Phase 1

Data analysis, feature engineering, Logistic Regression, XGBoost, evaluation and documentation

Future work only

Temporal deep learning, neural embeddings and rescheduling advice are not built yet

Dataset and Data Quality

Presenter: Data Analysis

336,776 rows

21 columns, 0 duplicate rows

9,430 rows (2.80%)

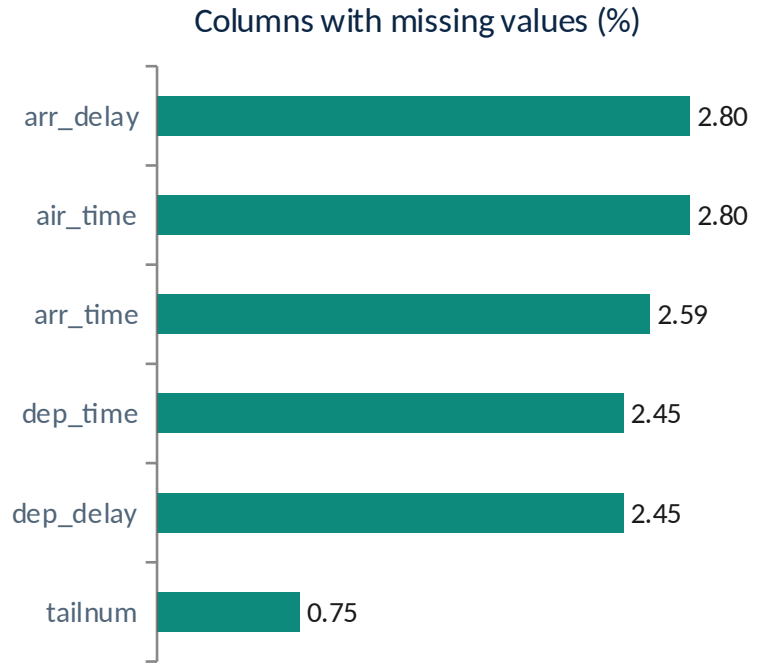
no arrival delay: removed

327,346 rows kept

processed file: 18 columns

3 origin airports

EWR, JFK, LGA



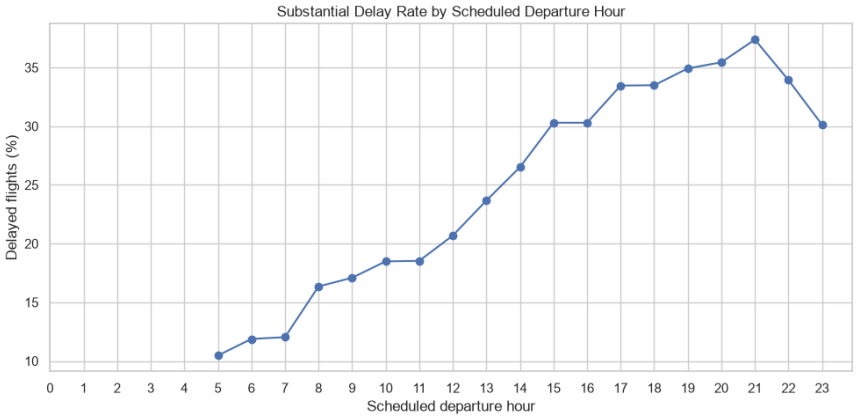
Source: notebooks/01_eda.ipynb

When Do Delays Happen?

Presenter: Data Analysis



Delay rate by month



Delay rate by scheduled departure hour

10% → 37%

delay rate from 5 am to 9 pm

33.6%

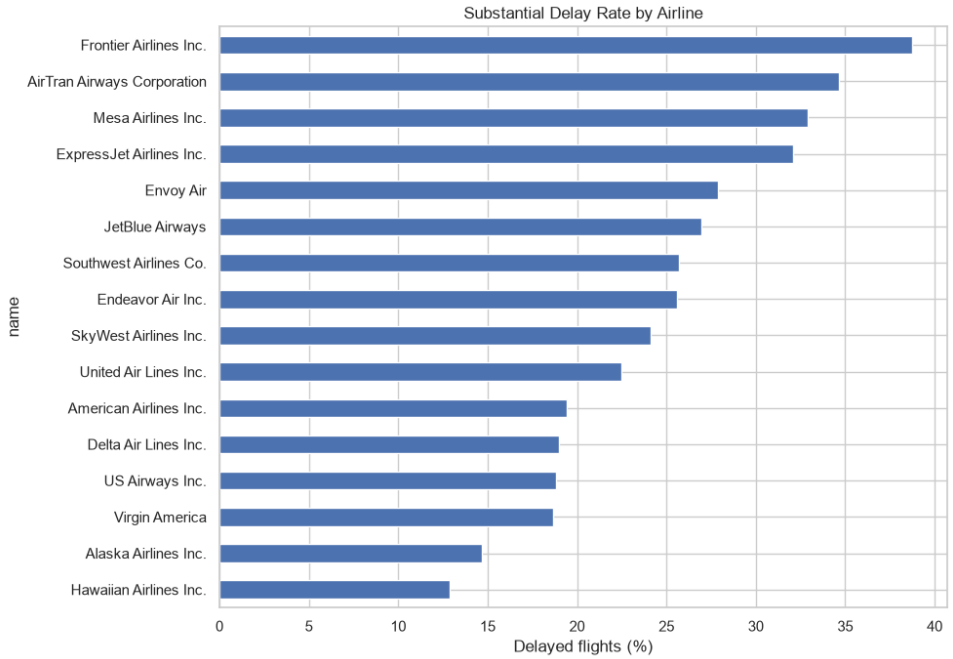
highest month: December

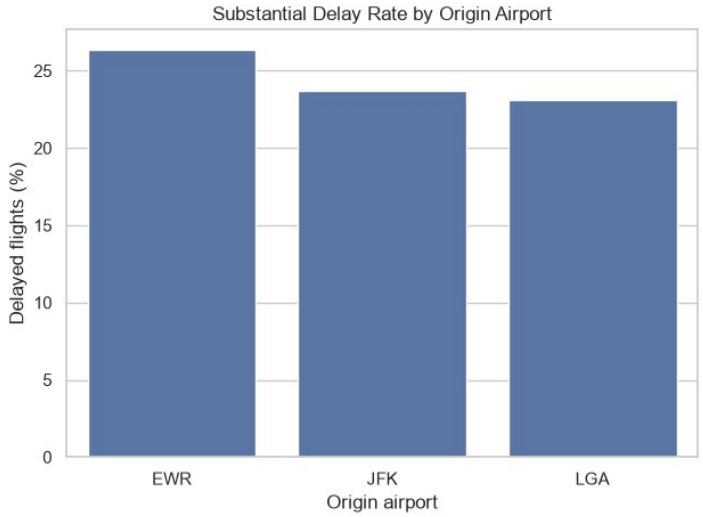
13.7%

lowest month: September

Delays by Airline and Airport

Presenter: Data Analysis



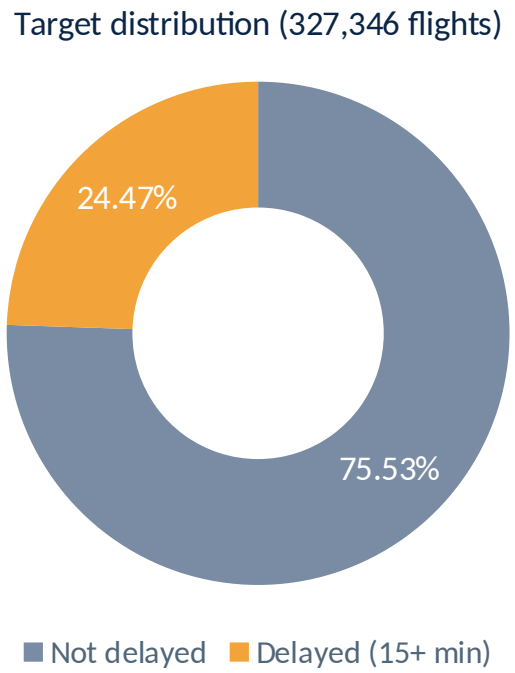


- Frontier 38.8% vs Alaska 14.7%
- Newark (EWR) 26.3% is highest
- Small airlines: few flights

Airline chart: notebooks/01_eda.ipynb

Target and Class Imbalance

Presenter: Feature Engineering



delayed = 1 if arr_delay ≥ 15 minutes, else 0

- 247,246 not delayed and 80,100 delayed
- About 1 in 4 flights is delayed
- Accuracy alone can mislead
- So we track PR-AUC, recall and F1
- Logistic Regression uses balanced class weights

16 Pre-Departure Features

Presenter: Feature Engineering

1

Calendar (4)

month, day, day of week, day of year

2

Clock time (2)

hour, minute

3

Cyclical (6)

sine and cosine of hour, day of week, day of year

4

Flight (4)

distance, carrier, origin, destination

Why sine and cosine? Hour 23 and hour 0 are close in time. The sine/cosine pair keeps this circular link.

Chronological Split, No Leakage

Presenter: Feature Engineering

Train · 70%	Validation · 15%	Test · 15%
229,142 flights	49,102	49,102
1 Jan – 13 Sep	13 Sep – 6 Nov	6 Nov – 31 Dec
delay rate 25.94%	**15.39%**	**26.69%**

Time flows from left to right: the model only learns from the past

Excluded (known only after flight)

dep_time, dep_delay, arr_time, arr_delay, air_time

Watch out

Validation delay rate (15.39%) is much lower than train and test, so thresholds may not transfer

Baseline: Logistic Regression

Presenter: Baseline Model

Scale numeric features → One-hot encode carrier, origin, dest → Logistic Regression balanced weights

0.2049
validation PR-AUC

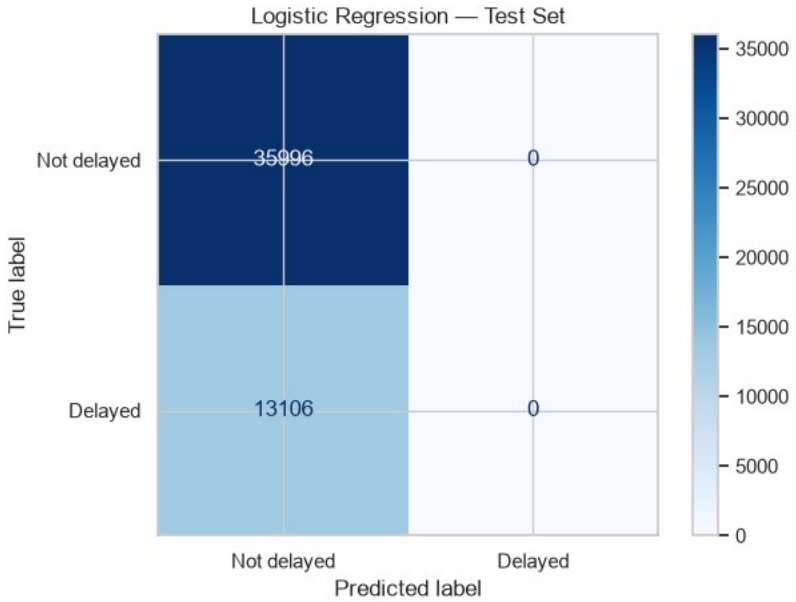
0.6105
validation ROC-AUC

0.7223
saved threshold

Threshold rule: highest recall with precision ≥ 0.60 on validation. Result: precision 0.667 but recall only 0.0003.

Logistic Regression Test Result

Presenter: Baseline Model



Confusion matrix on the test set (49,102 flights)

Metric	Logistic Regression
PR-AUC	0.308
ROC-AUC	0.568
Precision	0.000
Recall	0.000
F1-score	0.000
Accuracy	0.733

No positive predictions at the saved threshold. Accuracy 0.733 equals "always not delayed", so it is not a sign of skill.

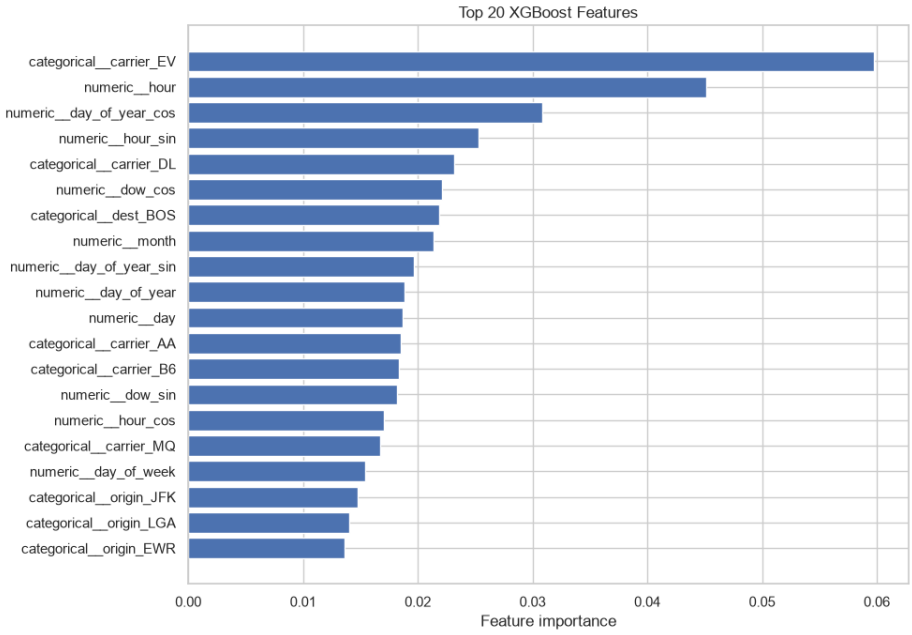
XGBoost: Gradient Boosting

Presenter: XGBoost

Setting	Value
Trees (n_estimators)	400
Maximum depth	6
Learning rate	0.05
Row subsample	0.8
Column subsample	0.8
Features / split	same as baseline

0.2405
validation PR-AUC

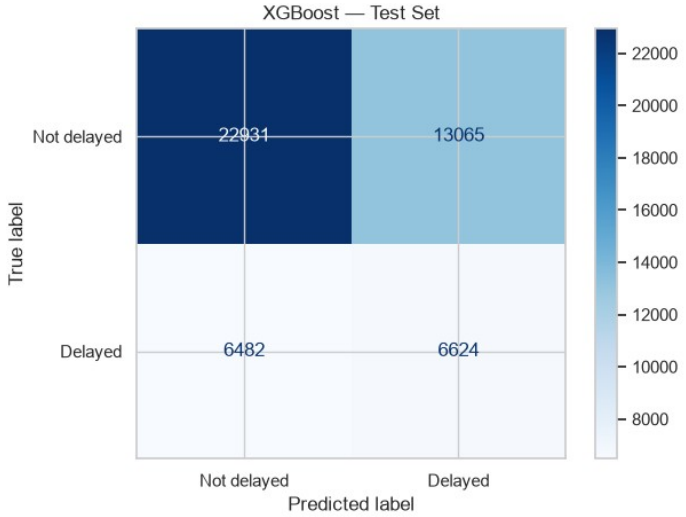
0.6371
validation ROC-AUC



Top 20 XGBoost features (notebook 04). Importance is not cause.

XGBoost Test Result and Threshold

Presenter: XGBoost



Test confusion matrix at threshold 0.30

0.336
precision

0.505
recall

0.404
F1-score

Validation threshold sweep

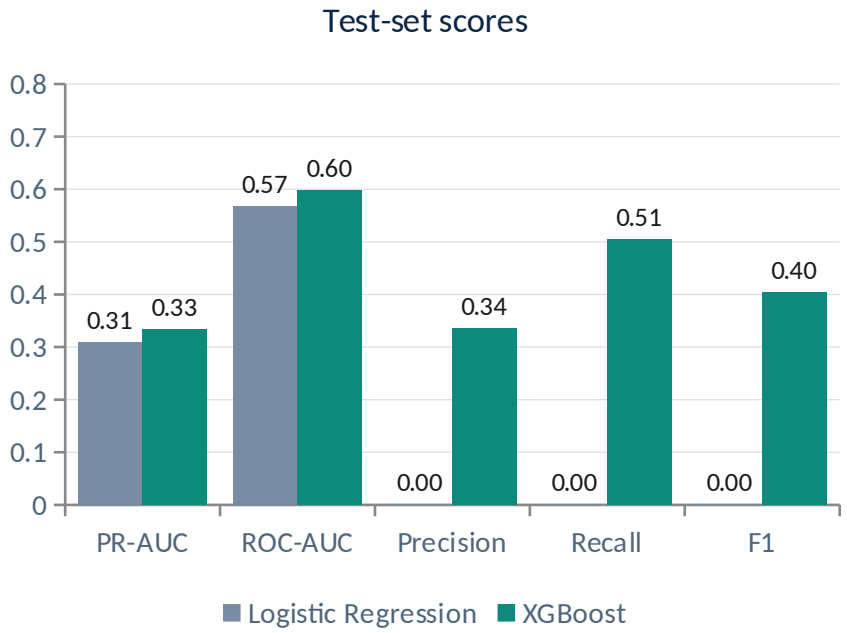
Threshold	Precision	Recall
0.2	0.200	0.703
0.3	0.218	0.564
0.4	0.236	0.428
0.5	0.256	0.290
0.7	0.333	0.083

The 0.30 threshold was set manually. On validation, the "precision ≥ 0.60" rule gave 0.8664 with recall of only 0.0016.

Model Comparison on Test Set

Presenter: Evaluation & Docs

Metric	Logistic Reg.	XGBoost
Threshold	0.722	**0.300**
PR-AUC	0.308	**0.334**
ROC-AUC	0.568	**0.598**
Precision	0.000	**0.336**
Recall	0.000	**0.505**
F1-score	0.000	**0.404**
Accuracy	0.733	**0.602**
Brier score	0.2092	**0.2091**

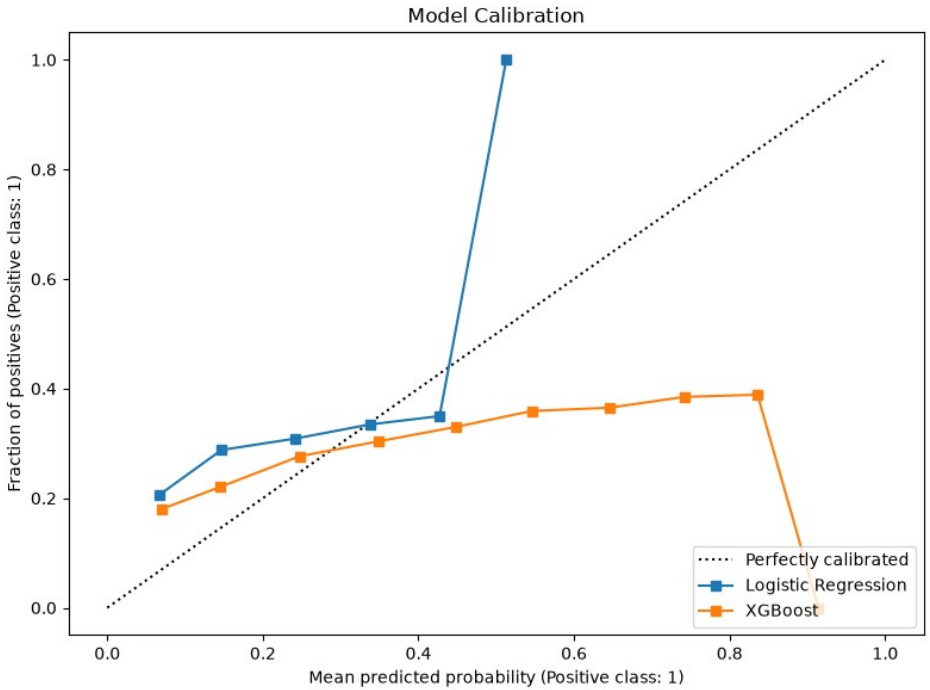


LR accuracy is high only because it never predicts a delay.

Source: reports/metrics/model_comparison.csv

Probability Quality (Calibration)

Presenter: Evaluation & Docs



0.2092
Brier: Logistic Reg.

0.2091
Brier: XGBoost

- Dotted line = perfect calibration
- XGBoost is over-confident at high probabilities
- Brier scores are almost identical
- Do not read probabilities as exact chances

Limitations and Honest Notes

Presenter: Baseline Model

1

Weak separation

ROC-AUC only 0.568 and 0.598

2

Schedule features only

No weather or earlier-flight delay data

3

One year, NYC only

2013 data may not match today

4

Validation mismatch

Delay rate 15.39% vs 25.94% / 26.69%

5

Threshold sensitivity

LR gives zero positives at 0.7223

6

Not for real use yet

Needs tuning and new-data testing

Presenter: XGBoost

Conclusion and Future Work

What Phase 1 shows

- A full pipeline: EDA, features, split, two models, evaluation
- XGBoost beats Logistic Regression on PR-AUC, ROC-AUC and recall at the reported thresholds
- Both models have limited discrimination
- Results are a baseline, not ready for real use

Future work (not implemented)

- Temporal deep learning models
- Neural-network embeddings
- Rescheduling recommendations
- More features, e.g. weather
- Better threshold tuning and calibration
- Test on newer data

Thank you. Questions?